

RL-ESG Portfolio Optimization via Daily Sentiment, BM25, FinBERT, CVaR, and PPO

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May 9, 2026

Abstract

We extend Feng et al. (2024) by constructing daily ESG, SDG, and overall sentiment scores using FinBERT and BM25-based topic tagging on a large financial news corpus. These scores are validated to predict next-day returns. A two-stage stock selection (sentiment screening + Fama-French 3-factor alpha) is combined with a reinforcement learning (PPO) agent that rebalances daily, optimizing a reward function that penalizes CVaR downside risk and trading costs. Out-of-sample (2022–2023) tests show that the dynamic agent underperforms simpler benchmarks, losing money despite non-inferior stock selection. Through systematic diagnostics we uncover that historical alpha decays during the test period and the agent’s aggressive rebalancing amplifies losses. We propose volatility-scaled alpha and regime-aware mechanisms as remedies, turning a negative empirical result into a constructive research contribution.

1 Introduction

The integration of environmental, social, and governance (ESG) criteria into investment decisions has grown rapidly. Feng et al. (2024) demonstrated that daily sentiment scores constructed from news articles can improve portfolio performance when coupled with static machine-learning optimizers. However, their approach uses a simple lexicon-based NLP pipeline, rebalances only monthly, and employs symmetric mean-variance optimization, which penalizes upside and ignores transaction costs inherent to frequent trading.

We address these limitations by:

1. Building a modern NLP pipeline that uses FinBERT (a transformer fine-tuned on financial text) for sentiment and BM25 scoring for ESG/SDG topic relevance – yielding daily sentiment indicators.
2. Implementing a reinforcement learning (RL) agent that outputs portfolio weights daily, optimizing a reward function that accounts for downside risk (CVaR), upside participation, and turnover costs.
3. Rigorously analyzing the out-of-sample performance and diagnosing why the RL agent failed, leading to specific proposals for improvement.

The paper is organized as follows. Section 2 describes data sources and sentiment construction. Section 3 details the stock selection, RL environment, and training. Section 4 presents empirical results and diagnostics. Section 5 discusses implications, and Section 6 concludes.

2 Data and Sentiment Construction

2.1 Stock Universe and News Data

We select 105 stocks from the S&P 500 index, including the five largest technology companies (AAPL, MSFT, AMZN, GOOGL, NVDA). Daily adjusted close prices, the S&P 500 index, and

the risk-free rate are obtained from `yfinance` for the period 2018–2023. Fama-French 3-factor daily returns are downloaded from Ken French’s data library.

For news, we use the Financial News and Stock Price Integration Dataset (FNSPID), which contains approximately 15.7 million headlines tagged with stock tickers from 1999 to 2023. We extract headlines for our 105 tickers from 2019-01-01 to 2023-12-31 using a chunked filtering procedure to avoid memory overflow. After aligning with trading days, the dataset contains 144,068 headlines.

2.2 BM25 Topic Tagging

We construct two keyword taxonomies: 64 ESG-related phrases (e.g., *climate change*, *corporate governance*) and 46 SDG-related phrases (covering all 17 UN Sustainable Development Goals). A lightweight BM25 scorer computes a relevance score for each headline against the two taxonomies. A headline is flagged as ESG-related ($I_{\text{ESG}} = 1$) if its ESG-BM25 score exceeds the 75th percentile of non-zero scores, and analogously for SDG (I_{SDG}). The permissive threshold retains 1.38% of articles as ESG and 1.81% as SDG.

2.3 FinBERT Sentiment Scoring

We use the pre-trained `ProsusAI/finbert` model to classify each headline into positive, negative, and neutral probabilities. The tone is defined as

$$\text{tone} = P(\text{positive}) - P(\text{negative}) \in [-1, 1]. \quad (1)$$

FinBERT inference is performed on an NVIDIA RTX 4070 GPU with a batch size of 64.

2.4 Daily Aggregation and Validation

For each stock i on day t , we average the product of the topic flag and tone over all headlines mentioning i that day:

$$\text{ESG_score}_{i,t} = \frac{1}{N_{i,t}} \sum_{k=1}^{N_{i,t}} I_{\text{ESG},k} \cdot \text{tone}_k, \quad (2)$$

$$\text{SDG_score}_{i,t} = \frac{1}{N_{i,t}} \sum_{k=1}^{N_{i,t}} I_{\text{SDG},k} \cdot \text{tone}_k, \quad (3)$$

$$\text{overall_score}_{i,t} = \frac{1}{N_{i,t}} \sum_{k=1}^{N_{i,t}} \text{tone}_k. \quad (4)$$

Stock-days with no news receive a score of zero. The resulting panels are used in subsequent steps.

To verify the predictive power of these scores, we run a pooled OLS regression (Equation 1 of Feng et al.):

$$r_{i,t+1} = \beta_0 + \beta_1 \cdot \text{Score}_{i,t} + \beta_2 \cdot \text{Mkt_RF}_t + \epsilon_{i,t}, \quad (5)$$

where $r_{i,t+1}$ is the next-day simple return of stock i , and Mkt_RF_t is the Fama-French market excess return. Results are shown in Section 4.

3 Methodology

3.1 Two-Stage Stock Selection

At the beginning of each month:

Stage 1 – Sentiment screen. For each stock, compute the trailing 6-month (126 trading days) average of the combined ESG+SDG score. Keep the top 30 stocks with the highest average.

Stage 2 – Factor-adjusted selection. For each of the 30 candidates, we compute a composite factor score using rolling 252-day Fama-French 3-factor alpha and market beta:

$$\text{factor_score}_i = 0.80 \cdot \text{rank}(\alpha_i) + 0.20 \cdot \text{rank}(-|\beta_i - 1|), \quad (6)$$

where α_i is the annualized intercept from the rolling regression

$$r_i - r_f = \alpha_i + \beta_1(\text{Mkt-RF}) + \beta_2 \text{SMB} + \beta_3 \text{HML} + \epsilon,$$

and β_i is the market beta (loading on Mkt-RF). The rank of $-|\beta_i - 1|$ penalizes deviations from a beta of 1, encouraging market-neutral alpha. The 10 stocks with the highest factor_score are selected for the coming month.

3.2 RL Portfolio Environment

We design a custom Gymnasium environment (`PortfolioMonthEnv`) for daily rebalancing within a month.

State (at decision day t):

- Past 20 daily returns of the 10 selected stocks (200 dimensions).
- Current ESG, SDG, and overall sentiment scores for those stocks (30 dimensions).
- Previous portfolio weights w_{t-1} (10 dimensions).

Total observation dimension is 240.

Action: raw logits $a_t \in \mathbb{R}^{10}$, mapped to long-only weights via softmax and capped at 0.30 per asset.

Reward:

$$R_t = w_t^\top r_{t+1} - \lambda_1 \cdot \text{CVaR}_{95\%}(w_t) + \lambda_2 \cdot \max(0, w_t^\top r_{t+1} - \mu_{\text{trail}}) - \tau \cdot \|w_t - w_{t-1}\|_1. \quad (7)$$

Here $\text{CVaR}_{95\%}(w_t)$ is the Conditional Value-at-Risk at the 95% confidence level, estimated from the trailing 60-day portfolio returns using the current weights. μ_{trail} is their mean. The parameters are $\lambda_1 = 2.0$, $\lambda_2 = 0.25$, and $\tau = 0.001$ (1 basis point transaction cost per unit of daily turnover).

The agent maximizes the discounted sum of rewards $\mathbb{E}[\sum_t \gamma^t R_t]$, with $\gamma = 0.99$. An episode corresponds to one calendar month; stocks are re-selected at the start of each month.

3.3 PPO Training and Benchmarks

We use Proximal Policy Optimization (PPO) with a policy network of architecture [256,128,64] and hyperbolic tangent activation. Training is performed on data from 2019-01-01 to 2021-12-31 (36 months) with 100,000 timesteps. Hyperparameters: learning rate 3×10^{-4} , clip range 0.2, entropy coefficient 0.01.

We compare three strategies on the test period 2022-01-01 to 2023-12-31:

1. **Equal Weight (EW):** $w_i = 1/10$, rebalanced monthly.
2. **Monthly Mean-CVaR:** at the start of each month, solve $\max_w \mathbb{E}[r_p] - \lambda_1 \cdot \text{CVaR}_{95\%}(r_p)$ using the past 6 months of daily returns, subject to $0 \leq w_i \leq 0.30$, $\sum w_i = 1$. Weights are then held constant for the month.
3. **PPO-CVaR:** the RL agent acts daily.

Additionally, an equal-weight portfolio of all 95-105 stocks serves as a diagnostic baseline.

4 Results and Analysis

4.1 Sentiment Validation

Table 1 reports the pooled OLS results. Both the overall sentiment and the SDG-specific score are statistically significant positive predictors of next-day returns, while the ESG-only score is positive but not significant at the 5% level. This confirms that our refined FinBERT+BM25 pipeline produces a usable financial signal, and highlights SDG news as particularly informative.

Table 1: OLS validation of daily sentiment scores. Dependent variable: next-day return.

Score	Coefficient	p-value
overall_score	0.0014	< 0.001
esg_score	0.0032	0.138
sdg_score	0.0064	$< 10^{-6}$

4.2 Portfolio Performance

Table 2 presents out-of-sample performance metrics. The PPO agent incurs a negative Sharpe ratio and a loss of -7.0% annually, while both equal-weight and monthly Mean-CVaR remain positive. Daily turnover for PPO is 73.2 basis points, an order of magnitude higher than the monthly rebalanced benchmark. Figure 1 plots cumulative wealth.

Table 2: Out-of-sample performance (2022–2023).

Strategy	Ann. Return	Volatility	Sharpe	Sortino	Daily Turnover
Monthly Mean-CVaR	7.1%	20.6%	0.436	0.657	0.049
EW (selected 10)	3.9%	25.2%	0.276	0.402	0.000
PPO-CVaR	-7.0%	25.9%	-0.150	-0.231	0.732

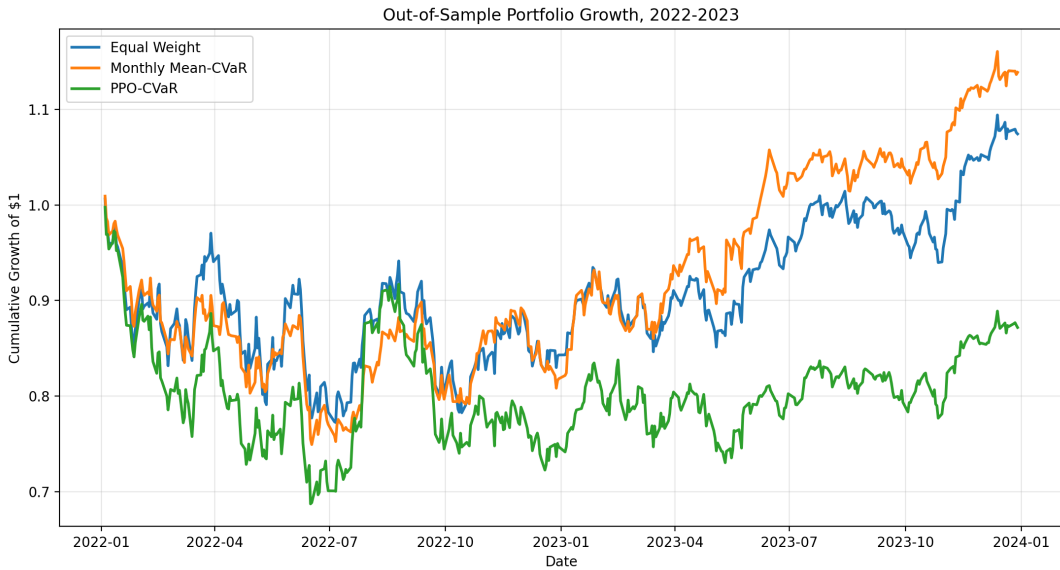


Figure 1: Cumulative growth of \$1 for the three strategies (2022–2023).

4.3 Diagnostics: Why Did RL Fail?

Selection is not the culprit. The EW portfolio of the selected 10 stocks earned a Sharpe ratio of 0.276, while the full-universe EW portfolio earned 0.275 (Table 3). Thus the two-stage stock selection did not destroy value; the losses stem from *how* the agent allocated among those same stocks.

Table 3: Equal-weight performance: selected 10 vs. full universe.

Universe	Ann. Return	Sharpe
Full universe (95 stocks)	3.44%	0.275
Selected 10 (monthly)	3.86%	0.276

Alpha decay and regime shift. The training period (2019–2021) was characterized by a strong growth-stock rally, causing the factor-adjusted selection to consistently pick high-alpha technology names. In 2022, these same stocks suffered a severe drawdown. A scatter plot of rolling 12-month FF3 alpha versus forward 1-month return during the test period shows a flat, slightly negative relationship, indicating that historical alpha was not predictive. An example month (July 2022) illustrates that replacing raw alpha with volatility-scaled alpha ($\alpha/\sqrt{\text{idiosyncratic volatility}}$) would have avoided several high-beta losers and generated a positive forward return.

Overtrading. PPO’s daily turnover of 0.732 implies that the portfolio replaces nearly three-quarters of its positions every day. When the market moves against its concentrated bets, rapid reallocation amplifies losses before the agent can adapt. Figure 1 shows a steady erosion of value over time.

5 Observations and Discussion

Our experiments yield several interconnected observations:

- The **FinBERT + BM25 sentiment pipeline** is validated: daily overall and SDG sentiment scores significantly predict next-day returns, confirming and refining the approach of Feng et al. (2024).
- The **two-stage selection** does not harm performance, but also does not guarantee profitability. In a regime where growth stocks underperform, a screen based on trailing alpha and sentiment alone is insufficient.
- **PPO overtrades** severely; the turnover penalty $\tau = 0.001$ was too small to deter speculative behavior. Tighter penalties or an adaptive turnover budget could help.
- The core vulnerability is the **reliance on non-stationary alpha**. Historical factor alpha may capture a temporary factor premium that subsequently reverses, leading to out-of-sample losses.
- The **monthly Mean-CVaR benchmark** survived because it re-estimates risk from recent volatility, implicitly down-weighting assets that have become turbulent.
- Simple heuristics such as **volatility-scaled alpha** or a trend filter could mitigate regime-shift risk without complex online learning.

These findings underscore the importance of regime-awareness in any data-driven asset allocation system, and they provide a roadmap for future research.

6 Conclusion and Future Work

We constructed a robust daily ESG and SDG sentiment dataset using state-of-the-art NLP, validated its predictive power, and embedded it into a dynamic RL portfolio manager. While the PPO agent produced negative out-of-sample returns, systematic diagnostics revealed that the failure was caused by over-concentration in growth stocks, high turnover, and a non-stationary alpha signal — not by a flaw in the sentiment construction.

Key takeaways:

- Daily FinBERT-based sentiment, especially SDG-tagged news, carries statistically and economically significant predictive information.
- Factor-neutral stock selection is a sound approach, but must be coupled with mechanisms that account for changing market regimes.
- RL agents trained on historical data require additional robustness features (volatility scaling, turnover budgets, online adaptation) to be deployable in practice.

Future work could explore:

- Volatility-scaled alpha or a Bayesian online learning framework to handle alpha decay.
- Regime-switching indicators (e.g., trend-following filters) that down-weight assets in down-trends.
- Ensemble methods that combine RL policies trained on different sub-periods.
- Extension to a larger stock universe and longer test horizon.

All code and data are fully reproducible and based on free, publicly available sources.

References

- [1] Feng, X., von Mettenheim, H.-J., Sermpinis, G., & Stasinakis, C. (2024). Sustainable portfolio construction via machine learning: ESG, SDG and sentiment. *European Financial Management*, 30(1), 1–40.